



Rajarata University of Sri Lanka

Faculty of Applied Sciences - Department of Computing

ICT 1402 – PRINCIPLES OF PROGRAM DESIGN AND PROGRAMMING Mid Term Examination 2021/2022 Batch

To be completed by the candidate

Examination Index No: _____ **Registration No:** _____

Important Instructions

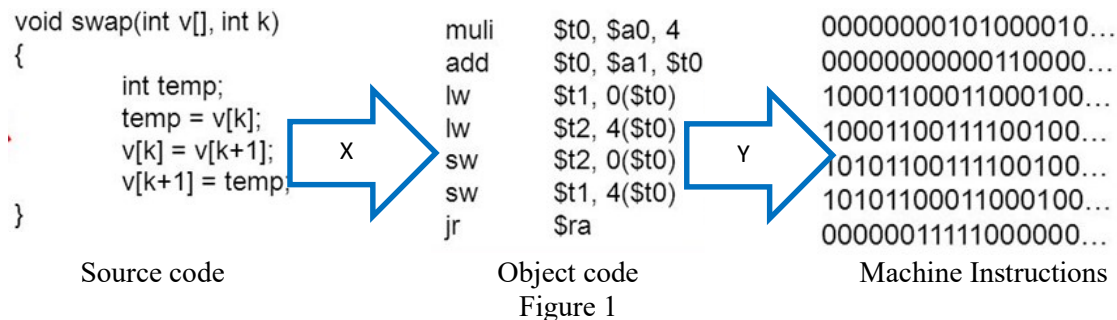
- The duration of the paper is **1 hour**.
- This paper contains 8 pages and **20** Multiple Choice Questions.
- Some question may have more than one correct answer. Select correct answer(s) and mark it in the answer script.
- **-0.25 Marks** will be given for each incorrect choice.
- Note that questions appear on both sides of the paper.
- If a page is not printed, please inform the examiner immediately.

1. Which of the following represent data and information respectively?
 - a. Number of emails received and the trend analysis of email communication.
 - b. GPS coordinates of a location and a detailed map with directions.
 - c. Ages of students in a class and the grade point average.
 - d. Rainfall measurements in millimeters and a weather forecast for the week.

2. Which of the following is/are true about Machine Language?
 - A. Each instruction in machine language corresponds to a specific operation that the computer can perform.
 - B. Machine language instructions are encoded using binary digits, representing low-level operations that a computer's central processing unit (CPU) can directly execute.
 - C. It is specific to the architecture of the computer.

a. A and B b. A only c. None of the above d. All of the above

3. Figure 1 depicts the translation of source code into machine code. What is the correct pair of language translators used in X and Y, respectively?



- a. Compiler, Interpreter.
 - b. Compiler, assembler
 - c. Interpreter, linker
 - d. Debugger, linker
-
4. Select the correct choice for filling in the blank in the following statement. The Von Neumann architecture describes a general framework that a computer's _____ should follow.
 - a. Hardware, data, and programming
 - b. Hardware, and information
 - c. Hardware, algorithms, and data.
 - d. Hardware, and data.

5. Which of the following correctly pair types of software with their respective categories?
 - a. Windows Operating System - Application Software
 - b. Photoshop - System Software
 - c. Device Drivers - System Software
 - d. Email client – Application software

6. Which is the correct task list for the following problem. “Find the amount of paint needed to coat the walls of a square-shaped room.”
 - a. set length, width, height, paint_coverage, total_area, paint_liters as 0
 get lenth of the room as l
 get width of the room as w
 get height of the room as h
 compute total_area = 2*(l*h+w*h)
 compute paint_liters = total_area/ paint_coverage
 disply paint_liters

 - b. set length, width, height, paint_coverage, total_area, paint_liters as 0
 get lenth of the room as l
 get width of the room as w
 get height of the room as h
 compute total_area = 2*(l*w*h)
 compute paint_liters = total_area/ paint_coverage
 disply paint_liters

 - c. set length, width, height, paint_coverage, paint_liters as 0
 get lenth of the room as l
 get width of the room as w
 get height of the room as h
 compute paint_liters = l*w*h/ paint_coverage
 disply paint_liters

 - d. set length, width, height, paint_coverage, total_area, paint_liters as 0
 get lenth of the room as l
 get width of the room as w
 get height of the room as h
 compute total_area = (l*h+w*h)
 compute paint_liters = total_area/ paint_coverage
 disply paint_liters

7. Which of the following statement(s) is/are false regarding assembly languages?
- a. Slower execution compared to C programs.
 - b. efficient in terms of execution speed.
 - c. less storage.
 - d. User friendly.
8. Which option(s) represents the correct sequence of the main phases of the software development life cycle?
- a. Review specification, informal design, formal design, code, test, maintain the program.
 - b. Review specification, task list, flow chart, pseudocode, test, maintain the program.
 - c. Review specification, informal design, algorithm, test, maintain the program.
 - d. Review specification, informal design, formal design, maintain the program, test.
9. Which of the following statement(s) correctly defines the information about the loop designed to print the following pattern.

```
* * * * *  
* * * * *  
* * * * *  
* * * * *
```

- a. Loop control variable: c, initial value of the loop control variable is 1, loop control condition: $c \leq 4$, repetitive statement: print “* * * * *”, and $c=c+1$
 - b. Loop control variable: c, initial value of the loop control variable is 0, loop control condition: $c \leq 4$, repetitive statement: print “* * * * *”, and $c=c+1$
 - c. Loop control variable: c, initial value of the loop control variable is 0, loop control condition: $c < 4$, repetitive statement: print “* * * * *”, and $c=c+1$
 - d. Loop control variable: c, initial value of the loop control variable is 0, loop control condition: $c < 4$, repetitive statement: print “* * * * *”, and $c=c+1$
10. Why is testing programs important in software development?
- a. To increase the complexity of the code.
 - b. To introduce bugs and errors intentionally.
 - c. To verify that the software functions as intended and identify potential issues.
 - d. To speed up the development process by skipping validation.

11. Choose the correct statement(s) after analysing the task list given below.

Until the	student count is 0
end of	enter student gpa
student	increment the student count by 1
results list	

- a. It calculates the GPA for all the students in the student results list.
- b. It displays the GPA of all the students in the student results list.
- c. It adds the GPA of all the students into the system.
- d. It adds and displays the GPA of all students in the results list.

12. Which of the following is/are true when designing flowcharts?

- A. It is designed based on three general design structures namely sequence, selection, and iteration.
- B. Each structure has a single input flow line and multiple exit flow lines.
- C. Flow lines in the flow charts show the flow of control through the program.

- a. A and C b. A only c. B only d. None of the above

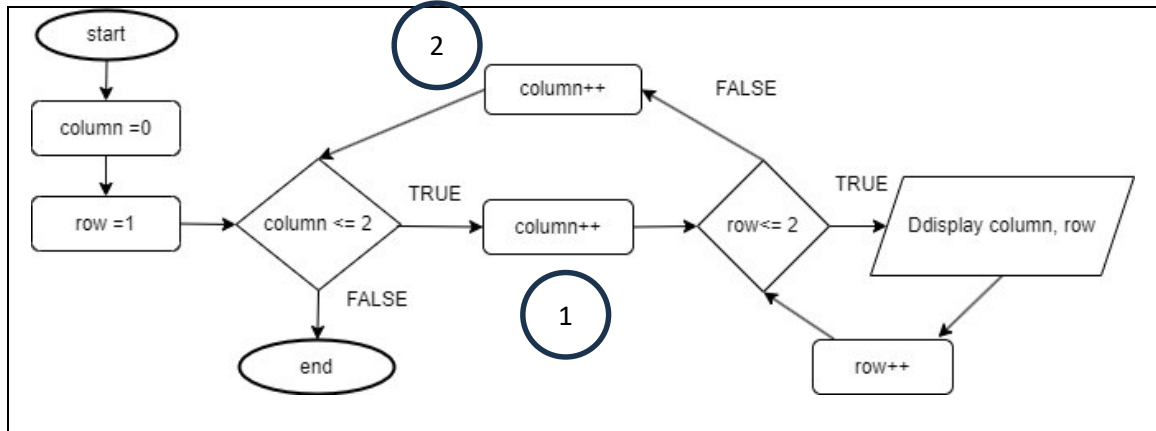
13. Which of the following statement(s) correctly explains loops?

- a. Until loop is the leading decision loop, and the while loop is a trailing decision loop.
- b. The body of the until loop will be executed once before control even gets to the loop condition.
- c. In until loop, the loop continues to repeat until the loop condition is true, whereas while loop continues to repeat if the loop condition is true.
- d. In the while loop, if the loop condition is false, control follows the first statement after the loop.

14. Which of the following best describes the distinction between internal processes and external processes in a flowchart?

- a. Internal processes represent interactions with external systems, while external processes depict the steps performed within the system.
- b. Internal processes illustrate the steps executed within the system, while external processes depict interactions with external entities or systems.
- c. Internal processes exclusively involve data storage, while external processes focus on data retrieval.
- d. Internal processes outline high-level tasks, while external processes detail low-level subroutines within the system.

15. Which statement(s) is/are true about the following flowchart?



- a. 1112
- b. 010211122122
- c. 010203
- d. 0102

16. How to modify the flow chart given in question 15 to obtain the following output?

0102

- a. Remove column++ process (labelled as 1) between the two conditions and add row=1 process in the false path of the second condition (row<=2).
- b. Remove column++ process (labelled as 2) between the two conditions and add row=1 process in the false path of the second condition (row<=2).
- c. Remove both processes labelled as 1 and 2.
- d. None of the above

17. Which statement(s) describes the valid rules followed when creating pseudocode?

- a. Write only one statement per line.
- b. Capitalize initial word.
- c. Indent to show hierarchy.
- d. Keep statements language independent.

18. What will be the output of the following pseudocode, if $n = 5$?

```
SET i and j = 1
READ n

WHILE i < n
    j = 1
    WHILE j < i
        DISPLAY "*"
        j = j + 1
    DISPLAY newline character
    i = i + 1
```

- a. * * * * *
- * * * * *
- * * * * *
- * * * * *
- b. * * * * *
- * * * * *
- * * * * *
- * * * * *
- c. *
- * *
- * * *
- d. *
- * *
- * * *

19. Choose the correct statement(s) from the following.

- a. Searching the student's marks ascending order is not an example for an algorithm.
- b. Third generation languages are also named as procedural languages.
- c. The body of the until loop will be executed once before control even gets to the loop condition.
- d. In the while loop, if the loop condition is true, control follows the first statement after the loop.

20. What is meant by a variable in programming?

- a. Variables are used to store information to be referenced and manipulated in a computer program.
- b. Variable is a keyword in C programming language.
- c. a. and b.
- d. none of the above

*** END ***